

DazzleVR: Enhancement of Brightness by Presenting Afterimage and Dazzle Reflex Sensation in Virtual Reality

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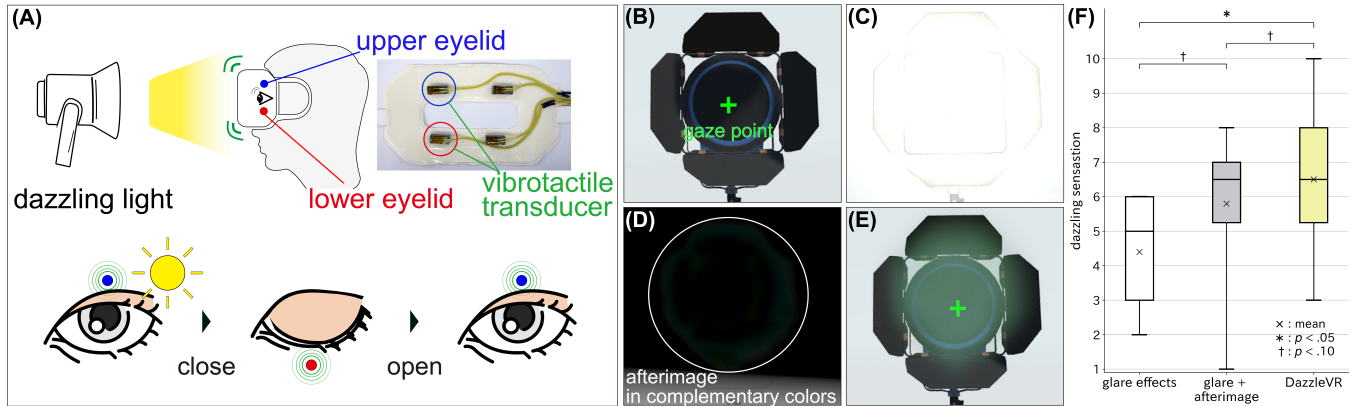


Figure 1: (A) Concept of pseudo dazzle reflex sensation by presenting haptic stimuli with a time lag (B) Light source in VR scene (C) Dazzling light with glare effects (D) Complementary color transitions of afterimage while blinking (E) Color and position transition of afterimage following gaze point (F) Perceived dazzling sensations under each condition

ABSTRACT

The presentation of brightness is essential in virtual reality. In this paper, we propose a system that can induce the sensation of dazzle by presenting a realistic afterimage and a pseudo dazzle reflex sensation with visuo-haptic feedback. Our system is expected to present realistic environments and emotional performance settings.

CCS CONCEPTS

• Human-centered computing → Virtual reality.

KEYWORDS

afterimage, blink reflex, brightness, dazzle reflex, virtual reality

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1 INTRODUCTION

Humans can perceive a wide range of luminosity, from modest light to glaring light, such as sunlight. The presentation of brightness is essential for reality in the visual experience. In the fields of computer graphics and virtual reality (VR), approaches utilizing developments in both hardware and software have been taken to present brightness more effectively. Approaches utilizing hardware include the head-mounted display (HMD) that can display a wide range of luminance [Matsuda et al. 2022]; however, physically reproducing high degrees of luminance can be problematic for heat dissipation, the size of the display, and the risk to the eyes. In approaches utilizing software, studies on capturing visual effects, including glare and afterimage presentation [E. Jacobs et al. 2015; Luidolt et al. 2020], have been conducted on standard 2D displays. However, apart from glare, research on capturing visual effects in VR has been limited. Additionally, the approaches presented in previous studies have yet to suggest a method for effectively reproducing the sensation of dazzling light, such as sunlight, which is difficult to observe directly.

Therefore, in addition to addressing the previously studied glare effects, we propose a novel approach to present the sensation of dazzle in VR by incorporating pseudo sensations, including afterimages and physical reflections, which occur after exposure to bright light. This study aims to show that the perceived light intensity is so high that it produces afterimage effects and physical reflections. To the best of our knowledge, this is the first study to enhance the

brightness using afterimage effects in VR and present the sensation of a dazzle reflex with multisensory stimuli.

2 IMPLEMENTATION

2.1 Presentation of realistic afterimage

When a person sees a dazzling light, an afterimage is generated in their field of vision and follows their gaze point. The color of the afterimage changes, and its intensity fluctuates before it eventually fades away over time. The phenomena of afterimage color transition and position shift based on eye-tracking data have already been implemented in 2D monitors [E. Jacobs et al. 2015]. We have extended the techniques underlying these developments to VR by applying them to the screen of each eye individually, presenting more realistic afterimages by newly implementing methods for fluctuating afterimage intensity and simulating complementary color transitions when one's eyelids are closed (Figure 1(D, E)).

2.2 Presentation of dazzle reflex sensation

When exposed to a dazzling light or air puff, one involuntarily experiences a blink reflex. Subsequently, while blinking one's eyelids, one may experience vision dimness, twitching, and stretching in the skin around the eyes. In addition, especially in the case of the dazzle reflex, afterimages are inverted to complementary colors while one's eyelids are closed [Berry 1922].

In this study, we aim to create the blinking sensation of the dazzle reflex in order to induce the perception of intense brightness that would lead to the blink reflex. We considered two methods for accomplishing this: one that actually provokes blinking, and one that reproduces a pseudo-blink sensation using visual and haptic stimuli. Regarding the former method, there is a way to produce blinking through electrical stimuli [Kono et al. 2018]. However, in this study, we present a pseudo-blinking sensation for ease of setup and to incorporate the presentation of a complementary color change of the afterimage during the blink. Therefore, the blackout and the afterimage transition during the blink were presented for visual feedback, while we presented skin vibrations in haptic feedback.

In the design of visual effects, to induce the perception that the blackout effect is caused by dazzling light, the duration of eyelid opening and closing was set to a reflex blink duration (opening: 125 ms, closing: 45 ms) [Evinger et al. 1991] and the afterimage of complementary colors was presented even during the blackout period. To present the skin vibration around the eyes, four transducers were placed on the face cushion, two on the upper eyelid and two on the lower eyelids (Figure 1(A)). The upper and lower transducers generated vibrations with a time lag corresponding to the opening and closing of the eyelids. The vibration waveform was set to a 30 Hz sinusoidal wave based on the findings of previous studies on around-eye vibration [Zeng et al. 2019].

3 PRELIMINARY USER STUDY

A preliminary user study was conducted to verify the effects of the proposed visuo-haptic feedback on the inducement of a pseudo dazzle reflex sensation and the extent to which the proposed methods can enhance the dazzling sensation. We established the following three conditions: exposure to light with only visual glare effects, exposure to light with visual glare and gaze-dependent afterimage

effects, and exposure to light under the DazzleVR system, presenting visual glare, afterimage effects and a dazzle reflex sensation with haptic stimuli. Ten volunteers experienced a VR scene in which they gazed at a light and gave a response about the dazzling sensation they experienced from the light (Figure 1(B–E)) on a 10-point scale (1: not dazzled at all, 10: extremely dazzled). The order of the exposure to the experimental conditions was randomized among the participants. After exposure to all the conditions, the participants answered a post-questionnaire about the dazzle reflex sensation and an open-ended question about their overall feelings.

Figure 1(F) shows the responses regarding dazzling sensations under each condition. Wilcoxon signed-rank tests performed under the Benjamini–Hochberg procedure showed that incorporating the pseudo dazzle reflex sensation in addition to the afterimage effects significantly improved dazzling sensation compared to the existing method that solely utilized the glare effect. In the open-ended answers, the following comments were made: "The feeling of the flickering afterimage after the blinding light was realistic" and "The light made me feel like blinking again and again." Additionally, six out of ten participants mentioned that the dazzle reflex sensations were evoked in the DazzleVR condition. These results suggest that the proposed visuo-haptic feedback system may induce the pseudo dazzle reflex sensation and that the presentation of the realistic afterimage and the pseudo dazzle reflex sensations can enhance the resulting sensation of dazzle, compared to existing methods.

4 USER EXPERIENCE

Our proposed methods can evoke the dazzling sensations induced by the sparkle of sunlight, the reflections of a snowfield, or a glimmer in the ocean, and can improve the sense of presence in the VR environment, making it useful for creating realistic VR sight-seeing applications. In addition, our system can exhibit dazzling illumination in performance settings, leading to enriched emotional experiences, and is expected to be used in the entertainment industry. At SIGGRAPH Asia 2023 Emerging Technologies, we plan to exhibit two experiences: a realistic setting of a forest with dazzling sunshine, and an exciting music concert with dazzling spotlights.

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